MILITARY SPECIFICATION

PLIERS, DIAGONAL CUTTING WITH CUSHION THROAT

1. SCOPE

1.1 This specification covers 4-1/2- and 7-1/2-inch size diagonal cutting pliers with cushion throat for cutting wire and cotter pins close to flat surfaces.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
GGG-P-471 - Pliers.

MILITARY
MIL-S-3136 - Standard Test Fluids Hydrocarbon and Iso-Octane.
MIL-H-5606 - Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance.
MIL-P-15424 - Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

FSC 5110

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3. REQUIREMENTS

3.1 General. - Pliers shall conform to the general requirements of Specification GGG-P-471, as applicable, and the following requirements:

3.1.1 Design. - Pliers shall be similar to type IV, class 1, style B pliers of Specification GGG-P-471 (see figure 1). Prior to the application of the plastic, the pliers shall conform to the dimensions shown in table I for the size specified (see 6.1).

Table I - Pliers, diagonal cutting with cushion throat.

Nominal size	A (inches)		B (inches)		C (inches)		D (inches)		E ±1/8	Jaw opening, minimum	Weight, maximum
	Dimensions	Tolerances	Dimensions	Tolerances	Dimensions	Tolerances	Dimensions	Tolerances			
Inches		Plus or minus		Plus or minus		Plus or minus		Plus or minus	Inches	Inch	Ounces
4-1/2	4-5/8	1/8	5/8	1/8	5/16	1/16	1/2	1/8	2	1/4	4
7-1/2	7-7/16	5/16	1	1/8	1/2	1/16	7/8	1/8	1-7/8	1/2	12

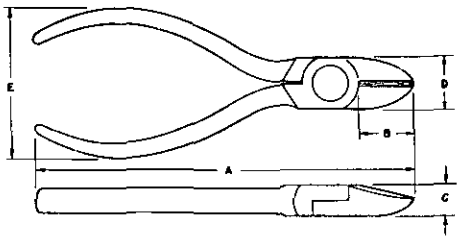


Figure 1 - Pliers, diagonal cutting with cushion throat.

3.1.2 Dimensions of throat surface. - The throat surfaces of the pliers shall be free from all oxide coating resulting from heat treatment. Each throat surface shall be recessed with grooves as follows: 4-1/2-inch pliers shall have a minimum of 3 grooves (forming at least 4 resultant ribs) and 7-1/2-inch pliers shall have a minimum of 5 grooves, (forming at least 6 resultant ribs).

3.1.2.1 Each groove shall extend from the outer edge of the throat to within 1/8-inch of the cutting edge on 4-1/2-inch pliers and to within 5/32-inch of the cutting edge on 7-1/2-inch pliers.

3.1.2.2 The depth of the groove closest to the joint shall be a minimum of 0.050 inch at a point approximately 1/8-inch from the cutting edge on 4-1/2-inch pliers and 5/32-inch from the cutting edge on 7-1/2-inch pliers. (From these points back to the outside edges of the throat, and toward the tip, the depth of all grooves may vary to a minimum depth of 0.030 inch. All dimensions taken from the cutting edges are to be measured in a plane parallel with the bottom of the grooves.)

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3.1.2.3 Each groove shall have a minimum width (measured at the top of the groove) of 0.040 inch for 4-1/2-inch pliers and 0.050 inch for 7-1/2-inch pliers.

3.1.2.4 The width at the top of any groove shall not exceed its width at the bottom by more than 0.004 inch, except that a 1/64 inch radius fillet will be allowable in the bottom corners.

3.1.2.5 The grooves and ribs shall be spaced at approximately uniform distances over a minimum area of 80 percent of the throat surfaces to which the plastisol is to be bonded. (When calculating this area, exclude the surface between the inside ends of the grooves and the cutting edges.)

3.1.3 Cushion material.- The cushion material shall be a vinyl plastisol and shall be properly primed for good adhesion to steel. Except for the relief area, the plastisol shall completely fill the throat of each jaw from within 1/16 inch of the tip to the joint end, and shall form an angle of approximately 90° between the cutting edge and the back side of the jaw. The plastisol shall be free from patches, cracks, holes, or any other defects which may adversely affect its serviceability.

3.1.3.1 The throat surfaces adjacent to the cutting edges shall be relatively free from plastisol along the entire length of the cutting edges to within 1/16 inch of the joint. This will provide a relief area into which the plastisol can be displaced when it is compressed under normal use. With the plier jaws closed, this area shall be large enough to allow a piece of drill rod or other round stock to be easily inserted through the entire length of the relief area. The diameter of the stock used for gaging shall be from 0.020 to 0.030 inch for the 4-1/2-inch pliers and 0.030 to 0.040 inch for the 7-1/2-inch pliers.

3.2 Marking.- Pliers shall be marked with the symbol "U.S." and the manufacturer's identification in accordance with Standard MIL-STD-130.

3.3 Workmanship.- Workmanship shall be first class in every respect. Heads, jaws, cutter throats, and handles shall be free from burrs and any projections which might cause injury to the hands of the operator. Pliers shall be free from rust and defects which may affect their serviceability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Sampling for lot acceptance.- Sampling for lot acceptance shall be performed as specified in Specification GGG-P-471.

4.3 Inspection.- Inspection shall be performed as specified in Specification GGG-P-471. Lot acceptance inspection shall consist of the tests specified in 4.4.

4.4 Test procedures.-

4.4.1 Hardness.- Hardness test shall be performed as specified in Specification GGG-P-471.

4.4.2 Handle-load test.- Handle-load test shall be performed as specified in Specification GGG-P-471.

4.4.2.1 Allowable deflection and set.- Allowable deflection and set shall be determined in accordance with Specification GGG-P-471.

4.4.3 Cutting tests.- Steel-wire cutting test, lamp cord cutting test and paper cutting test shall be performed as specified in Specification GGG-P-471.

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4.4.4 Jaw opening test. - Jaw opening test shall be performed as specified in Specification GGG-P-471.

4.4.5 Plastisol tests. - The following plastisol tests (except for the solvent test) shall be performed after the plastisol inserts have been applied to the plier jaws:

4.4.5.1 Hardness test. - Plastisol shall show a hardness equal to a 10 second reading of 50 to 75 points on a Shore "A-2" durometer. If more convenient, a Rex hardness gage or equal may be used. (The Rex hardness gage is manufactured by the Naugatuck Chemical Division, U.S. Rubber Company, Naugatuck, Connecticut.)

4.4.5.2 Simulated aging test. - After being oven aged at 158° ±4° Fahrenheit (F.). for 48 hours, the sample shall be removed from the oven and allowed to cool to room temperature (65° to 75°F.). The hardness test shall again be performed and the increase in hardness shall be not more than 10 points over the hardness reading taken prior to this test.

4.4.5.3 Cutting tests. - The applicable cutting tests of Specification GGG-P-471 shall be performed only after the plastisol has been applied to the pliers and subjected to the tests specified above. During performance of each cutting test, the cushion throat jaws shall securely hold the cut-off piece of wire after each cut is made.

4.4.5.4 Resiliency test. - Pliers shall make three consecutive cuts of lamp cord as specified in Specification GGG-P-471. Cuts shall be made at a point approximately half way between the tip and the hinge end of the cutting edges. The three cuts shall be spaced approximately 5 to 10 seconds apart, and within 5 seconds after making the last cut of lamp cord, the pliers shall be made to cut 0.034 inch diameter steel wire at approximately the same point on the cutting edge. The plastic throat shall securely hold the cut-off piece of steel wire.

4.4.5.5 Visual examination. - Examination of the plastisol after each of the above tests shall show no adverse affect upon the plastisol, and no loosening of the plastisol from the jaws of the pliers.

4.4.5.6 Solvent test. - A test specimen of plastisol, having the same composition and properties as the plastisol used on pliers furnished under this specification, shall be fabricated for the solvent test. The specimen shall not exceed 1/2 cubic inch in volume and shall be fabricated and cured under the same conditions employed in application of plastisol to the pliers. The specimen shall be completely submerged in hydraulic oil in accordance with Specification MIL-H-5606 and hydrocarbon fluid in accordance with Specification MIL-S-3136 for a period of 24 hours in each fluid. The plastisol shall show no greater than three percent increase or one percent decrease in volume upon removal from the fluids.

4.5 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. -

5.1.1 Levels A, B, and C. - Pliers shall be protected for the level of shipment specified (see 6.1) in accordance with Specification MIL-P-15424.

5.2 Marking. - In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with Standard MIL-STD-129.

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6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of plier required (see 3.1.1).
(c) Whether preservation and packaging should be by level A or C, and whether packing should be by level A, B or C (see 5.1.1).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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